

UNIVERSITY OF ENGINEERING AND TECHNOLOGY



Nguyễn Hải Nam

**MULTI-AGENT VIRTUAL GROUP
THERAPY FOR STUDENT MENTAL
HEALTH ISSUES**

BACHELOR THESIS

Major : Artificial Intelligence

Hanoi, 2026

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ĐẠI HỌC CÔNG NGHỆ, ĐẠI HỌC QUỐC GIA HÀ NỘI

Nguyễn Hải Nam

HỆ THỐNG TRỊ LIỆU NHÓM ẢO ĐA TÁC
NHÂN HỖ TRỢ SỨC KHỎE TÂM LÝ SINH
VIÊN

KHÓA LUẬN TỐT NGHIỆP ĐẠI HỌC HỆ CHÍNH QUY

Ngành: Trí tuệ nhân tạo

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Hanoi, 2026

Authorship

I hereby declare that the thesis is my own original work, conducted under the supervision of Ph.D. Tran Van Khanh. All references to related research have been clearly cited in the bibliography. This thesis does not contain any material copied from others' works without proper acknowledgment.

Signature:.....

Supervisor's approval

"I hereby approve that the thesis in its current form is ready for committee examination as a requirement for the Artificial Intelligence degree at the University of Engineering and Technology."

Signature:.....

Acknowledgments

I would like to express my sincere gratitude to Ph.D. Tran Van Khanh for having guided and supported me throughout the process of carrying out this thesis. This valuable knowledge will be a foundation for my future career.

I would like to express my heartfelt appreciation to all lecturers at the Institution of Artificial Intelligence for their expert mentorship and dedication. Additionally, I am deeply thankful to my friends

Abstract

Students today frequently face psychological pressures but encounter significant barriers in accessing traditional mental health services. With the rapid development of Generative AI, they have the opportunity to access psychological support methods more deeply and easily by interacting with AI chatbots. However, most current systems act solely as one-on-one therapists, lacking the empathy and social interaction inherent in group therapy. This highlights a promising research direction: simulating a virtual group therapy space using a Multi-Agent System (MAS) based on large language models (LLMs). In this environment, agents act not only as therapeutic orchestrators but also as peers to share and empathize with the students. However, maintaining natural and safe interactions in a multi-participant virtual group is highly challenging. This thesis presents the design of agents based on Cognitive Behavioral Therapy (CBT), Behavioral Activation (BA), and Mindfulness-Based Intervention (MBI) principles, and proposes a shared state architecture (Blackboard Pattern) integrated with safety guardrails for coherent conversational management. Evaluation results indicate that the system effectively coordinates agent roles and proactively adapts responses to different clinical stages, fostering a safe, multidimensional, and effective psychological support environment.

Keywords: Generative AI, Student mental health, Virtual group therapy, Cognitive Behavioral Therapy (CBT), Behavioral Activation (BA), Mindfulness-Based Intervention (MBI), Multi-Agent System (MAS), Blackboard Architecture.

Tóm tắt đồ án

Học sinh, sinh viên hiện nay thường phải đối mặt với nhiều áp lực tâm lý nhưng lại gặp rào cản trong việc tiếp cận các dịch vụ hỗ trợ sức khỏe tinh thần truyền thống. Với sự phát triển mạnh mẽ của Generative AI, học sinh, sinh viên có cơ hội tiếp cận các phương pháp hỗ trợ tâm lý một cách sâu rộng và dễ dàng hơn thông qua việc tương tác với chatbot AI. Tuy nhiên, hầu hết các hệ thống hiện nay mới chỉ đóng vai trò như một nhà trị liệu trực tiếp 1-1, thiếu đi sự đồng cảm và tương tác xã hội vốn có của các buổi trị liệu nhóm. Điều này mở ra một hướng nghiên cứu tiềm năng: mô phỏng không gian trị liệu nhóm ảo đa tác tử (Multi-Agent System – MAS) dựa trên các mô hình ngôn ngữ lớn (LLMs). Trong môi trường này, các tác tử không chỉ đóng vai trò chuyên gia điều phối mà còn giả lập những người bạn đồng hành (peers) để chia sẻ và thấu cảm với sinh viên. Tuy nhiên, việc duy trì tính tự nhiên và an toàn trong một nhóm đa thành phần là một thách thức lớn. Đồ án này trình bày quy trình thiết kế các tác tử theo các nguyên tắc Trị liệu Nhận thức Hành vi (CBT), Kích hoạt Hành vi (BA) và Can thiệp dựa trên Chánh niệm (MBI), đồng thời đề xuất kiến trúc chia sẻ trạng thái chung (Blackboard Pattern) kết hợp cơ chế kiểm duyệt an toàn nhằm quản lý luồng hội thoại mạch lạc. Kết quả cho thấy hệ thống không chỉ phối hợp linh hoạt các vai trò mà còn chủ động phản hồi phù hợp với từng giai đoạn trị liệu, tạo ra một không gian hỗ trợ tâm lý đa chiều, an toàn và hiệu quả.

Từ khóa: Generative AI, Sức khỏe tinh thần học đường, Trị liệu nhóm ảo, Trị liệu Nhận thức Hành vi (CBT), Kích hoạt Hành vi (BA), Can thiệp dựa trên Chánh niệm (MBI), Hệ thống đa tác tử (MAS), Kiến trúc Blackboard.

Contents

ABSTRACT	ii
LIST OF FIGURES	viii
LIST OF TABLES	ix
1 INTRODUCTION	1
1.1 Background and Research Motivation	1
1.2 Research Problem and Gap	2
1.3 Research Objectives, Questions, and Scope	3
1.4 Thesis Structure	5
2 LITERATURE REVIEW	6
2.1 AI-Based Mental-Health Support	6
2.1.1 Screening Measures	6
2.1.2 LLM Support Systems	7
2.1.3 One-to-One Limits	8
2.2 Therapeutic Foundations for Structured Support	8
2.2.1 CBT Competence	8
2.2.2 MBI and BA	9
2.2.3 Safety Boundaries	10
2.3 Virtual Group Therapy	10
2.3.1 Why Group Support	10
2.3.2 Yalom Factors	11
2.3.3 Blackboard Coordination	12
2.4 Evaluation of LLM Mental-Health Systems	12
2.4.1 Behavioral Safety	12

2.4.2	Judges and Synthetic Data	13
2.4.3	VSM Requirements	14
2.5	Summary and Research Gap	14
3	METHODOLOGY	17
3.1	Design Goals	17
3.2	Therapist Core	19
3.2.1	Clinical Intake and Routing	20
3.2.1.1	Route Selection	20
3.2.1.2	Crisis Bypass	20
3.2.2	Evidence-Gated Stage Control	20
3.2.2.1	Evidence Extraction	20
3.2.2.2	Route Contracts	21
3.2.2.3	Stage FSM	22
3.2.3	Therapist Reasoning	22
3.2.3.1	Case Formulation	22
3.2.3.2	Stage Knowledge and Style	23
3.2.3.3	Therapist Planning	24
3.3	Blackboard Coordination	25
3.3.1	Shared State and Dialogue Graph	25
3.3.2	Output Assembly	26
3.3.3	Visible Memory and Internal Trace	26
3.4	Peer Support Layer	27
3.4.1	Yalom-Inspired Peer Design	27
3.4.1.1	Nam Peer Role	27
3.4.1.2	Linh Peer Role	27
3.4.2	Contribution Control	28
3.4.2.1	Contribution and Silence	28
3.4.2.2	Persona Validation	28
3.4.2.3	Therapist Approval	28
3.5	Safety and Evaluation Trace	29
3.5.1	Validation and Supervisor Audit	29
3.5.1.1	Response Validation	29
3.5.1.2	Supervisor Audit	30
3.5.2	Psychosocial Safety	30
3.5.2.1	Safety Critics	30

3.5.2.2	Crisis Bypass	30
3.5.3	Benchmark Logs	31
4	EXPERIMENTAL SETUP AND EVALUATION	33
4.1	Method	33
4.1.1	Tasks	33
4.1.2	LLM-as-a-Judge	35
4.1.3	Compared Systems and Score Reporting	35
4.2	Dataset	36
4.2.1	Dataset Design	36
4.2.2	Dataset Statistics	37
4.3	Main Results	38
4.3.1	Overall Performance	38
4.3.2	Route-Level Performance	39
4.3.3	Safety and Robustness	40
4.3.4	CAPE-II Competence Analysis	41
4.3.5	Yalom Group Dynamics Analysis	42
4.4	Analysis	43
4.5	Discussion	43
5	CONCLUSION	44
5.1	Summary of Contributions	44
5.2	Limitations	45
5.3	Ethical Boundaries and Future Work	45
5.4	Conclusion	46
	REFERENCES	49
	APPENDICES	50
A	ROUTE STAGE CONTRACTS	50
B	AGENT PROMPTING AND SAFETY BOUNDARIES	52
B.1	Therapist Orchestrator	52
B.2	Nam: Emotional Mirror	52
B.3	Linh: Hope Witness	53
B.4	Safety Boundary Statement	53

C	SYSTEM AND BENCHMARK ARTIFACTS	54
C.1	System Implementation Notes	54
C.2	Benchmark Artifacts	54
C.3	Recommended Final Evaluation Procedure	55

List of Figures

3.1.1 Therapist-led virtual group therapy architecture.	18
3.2.1 Evidence extraction and stage FSM pipeline.	21
3.2.2 Therapist core pipeline.	25
3.3.1 Blackboard coordination graph.	25
3.4.1 Peer contribution decision flow.	29
3.5.1 Safety and benchmark trace pipeline.	32
4.3.1 Overall VSM profile.	39
4.3.2 Route-level technique fidelity.	40
4.3.3 Safety and adversarial robustness diagnostics.	41

List of Tables

2.4.1 Representative related work and the design gap motivating this thesis.	15
2.5.1 Mapping from literature themes to system design decisions.	16
3.2.1 Evidence extraction and repair strategy.	21
3.2.2 Route contract fields used by the therapist core.	22
3.2.3 Route-stage summary for CBT, MBI, and BA.	23
3.2.4 Stage FSM transition policy.	23
3.2.5 Therapist planner input and output contract.	24
3.3.1 Main state groups in the blackboard.	26
3.4.1 Peer role contract.	28
3.4.2 Peer contribution and silence policy.	29
3.5.1 Supervisor audit checks.	30
3.5.2 Safety critics and fallback actions.	31
4.1.1 Compared systems in the full benchmark.	36
4.1.2 Reported task-related indicators.	36
4.2.1 Dataset split statistics for VSM-All.	37
4.2.2 Case-group composition in VSM-All.	38
4.3.1 Overall VSM-All leaderboard with comparative LLM judging. . . .	38
4.3.2 Route-level technique fidelity (%) by system.	40
4.3.3 Safety and robustness diagnostics across specific risk dimensions. .	41
4.3.4 CAPE-II inspired competence dimensions.	41
4.3.5 Yalom therapeutic-factor scores.	42
A.0.1CBT stage contract used by the prototype.	50
A.0.2MBI stage contract used by the prototype.	51
A.0.3BA stage contract used by the prototype.	51

Chapter 1

INTRODUCTION

1.1 Background and Research Motivation

University students face a combination of academic pressure, social comparison, family expectations, career uncertainty, loneliness, and burnout during a period when many mental-health difficulties first become visible. Large-scale studies of college students show that mental-health concerns are common in this population [1], while recent trend analyses report continuing concerns around student mental health and help-seeking [12]. At the same time, students may delay or avoid formal counseling because of stigma, limited access, cost, time constraints, or uncertainty about whether their distress is serious enough to discuss with a professional. Stigma and help-seeking barriers among college students have been documented empirically [5].

Digital support tools are therefore attractive because they can be low-barrier, always available, and easier to approach than formal services. Early conversational-agent work such as Woebot explored automated CBT-oriented support for young adults with symptoms of depression and anxiety [7]. More recent LLM-based systems such as SoulChat show that language models can be adapted toward empathetic listening, comfort, and client-centered dialogue [3]. These systems suggest that conversational AI can provide useful supportive interaction, especially when users need reflection, organization of thoughts, or encouragement to seek stronger real-world support.

However, mental-health dialogue is not simply a problem of producing warm text. A response can be fluent, empathetic, and still clinically poorly timed. For example, a user who is physiologically overwhelmed may need grounding before cognitive analysis; a user who has already identified an automatic thought may need guided discovery rather than another broad request to describe feelings. Work on cognitive reframing and CBT-oriented counseling conversations suggests that therapeutic technique and response timing matter [10, 18]. This thesis is motivated by the gap between fluent chatbot conversation and structured, inspectable, safety-aware supportive dialogue.

1.2 Research Problem and Gap

Many LLM-based support systems follow a one-to-one design: the user sends a message, and a single model produces a response. This design is simple, but it compresses many responsibilities into one component. The same model is expected to understand the user’s state, choose a therapeutic direction, maintain context, generate the response, and respect safety boundaries. In student mental-health conversations, this can lead to generic reassurance, repeated validation, premature advice, route mixing, or poor stage progression.

The first gap is a process-control gap. Existing LLM counseling systems can generate supportive responses, but it remains difficult to ensure that the conversation follows a coherent therapeutic route across multiple turns. A student-support system should be able to hold, advance, or regress a stage based on evidence in the conversation. It should also avoid mixing CBT, mindfulness-based intervention, and behavioral activation when the user’s current need points to only one of them. Behavioral assessment work argues that therapist-like systems should be evaluated through observable counseling behavior rather than general chatbot quality alone [4].

The second gap is a virtual-group gap. Human group therapy theory emphasizes factors such as universality, catharsis, hope, and interpersonal learning [23]. Yet most LLM mental-health systems remain one-to-one chatbots. Adding multiple agents without control would not solve this problem; it could make the dialogue repetitive, confusing, or unsafe. The open question is how to operationalize peer-like support so that virtual peers can contribute when useful while the therapist

remains the final responsible voice.

The third gap is an evaluation gap. If an evaluation only rewards fluency, empathy, or general helpfulness, it may miss whether the system followed the intended modality, advanced the conversation appropriately, used peer support at the right time, or handled safety boundaries. LLM-as-a-judge methods can support scalable evaluation, but they also require caution and calibration [25]. Mental-health counseling and psychosocial safety work further emphasizes the need to test boundary control, robustness, and sensitive-risk scenarios [11, 15]. This thesis therefore treats system design and benchmark design as connected problems.

1.3 Research Objectives, Questions, and Scope

The objective of this thesis is to design and evaluate a controllable AI support prototype for Vietnamese student mental-health conversations. The work focuses on dialogue structure, agent coordination, inspectability, and safety boundaries. It does not attempt to prove treatment outcomes, replace professional counseling, or deploy a clinical product.

The thesis is guided by three research questions:

1. **RQ1:** How can an LLM-based student-support system maintain route and stage control across CBT, mindfulness-based intervention, and behavioral activation?
2. **RQ2:** How can Yalom-inspired peer agents contribute to a virtual group-support experience without taking clinical authority away from the therapist?
3. **RQ3:** How can such a system be evaluated beyond response fluency, using safety, modality fidelity, group dynamics, context retention, and multi-turn progress?

To answer these questions, the thesis proposes a blackboard-based multi-agent virtual group therapy prototype. The conversation begins with onboarding and screening. DASS-21 is used as a practical screening signal for initial routing, supported by its original formulation and Vietnamese youth validation work [9, 14]. The system then routes the conversation into CBT, mindfulness-based intervention, or behavioral activation. CBT is included for cognitive restructuring and

guided discovery [8]; mindfulness-based intervention is included for grounding and decentering in high-arousal or ruminative states [24]; and behavioral activation is included for low-energy, avoidance, and academic-burnout patterns [16].

The system uses a shared blackboard state to coordinate specialized components. A clinical assessor updates route, stage, safety flags, and case formulation. Two peer agents, Nam and Linh, observe the blackboard and may draft short supportive contributions only when the current stage and Yalom factor allow it. Nam is designed for universality and catharsis; Linh is designed for hope and interpersonal learning. The therapist orchestrator remains the final decision maker: it reviews peer drafts, rewrites or discards them when needed, and produces the user-visible response. This design follows multi-agent and blackboard coordination principles while avoiding uncontrolled multi-speaker dialogue [17, 21].

The thesis makes four main contributions:

1. **A stage-constrained dialogue architecture** for CBT, mindfulness-based intervention, and behavioral activation, including route goals, transition logic, and response validation.
2. **A therapist-governed blackboard coordination mechanism** that separates assessment, peer observation, therapist planning, guardrails, and memory update through shared state.
3. **A Yalom-informed peer policy** in which Nam and Linh provide limited, therapist-approved support while preserving silence when peer contribution would harm the therapeutic flow.
4. **A VSM benchmark workflow** for multi-turn evaluation of safety, modality fidelity, group dynamics, conversation progress, reliability, confidence intervals, and human-audit readiness.

The scope of the thesis is deliberately limited. The system is a research prototype for supportive dialogue, not a diagnostic instrument or a substitute for licensed care. The benchmark is designed for engineering evaluation and does not establish treatment outcomes. Real-world use would require clinician review, stronger baseline comparison, privacy analysis, safety monitoring, and formal ethical evaluation.

1.4 Thesis Structure

The remainder of this thesis is organized as follows. Chapter 2 reviews screening measures, therapeutic foundations, LLM-based mental-health systems, Yalom-inspired group support, blackboard multi-agent coordination, and evaluation methods. It also identifies the process-control, virtual-group, and evaluation gaps addressed by this work.

Chapter 3 presents the methodology and system design. It describes the blackboard state, dialogue graph, clinical assessor, stage contracts, case formulation layer, peer agents, therapist orchestrator, guardrails, memory updater, and implementation interfaces.

Chapter 4 describes the experimental setup and evaluation framework. It introduces the VSM benchmark, dataset splits, compared systems, metrics, hybrid scoring, structural results, and staged evaluation plan.

Chapter 5 concludes the thesis by summarizing the contributions, discussing limitations and ethical considerations, and outlining future work for stronger evaluation and safer deployment.

Chapter 2

LITERATURE REVIEW

This chapter reviews the literature that shapes the design of the proposed system. It follows a funnel structure. The chapter first discusses AI-based mental-health support and why student-facing dialogue systems require more than fluent conversation. It then reviews therapeutic foundations that motivate route-specific structure for CBT, mindfulness-based intervention, and behavioral activation. The third section develops the idea of virtual group therapy, where Yalom-inspired peer support is combined with multi-agent blackboard coordination. The fourth section reviews evaluation and safety methods for LLM-based mental-health systems. The chapter closes by identifying the research gap addressed by this thesis.

2.1 AI-Based Mental-Health Support

2.1.1 Screening Measures

Student-support systems need an initial understanding of the user’s distress before deciding how to respond. In psychological and digital-health settings, this first signal is often gathered through short self-report screening instruments. The Depression Anxiety Stress Scales were introduced to measure symptoms associated with depression, anxiety, and stress [14].

This thesis uses DASS-21 because its three-factor structure is convenient for the prototype’s initial route selection. In this implementation, stress, anxiety,

and depressive-pattern signals are treated as approximate routing cues rather than clinical rules. DASS-21 is also suitable for this thesis because it has been validated within the Vietnamese student context [9]. Although using this instrument does not make the proposed chatbot clinically validated, it establishes DASS-21 as a practical screening reference for a domestic student-support prototype.

The system therefore treats screening as a routing aid, not as diagnosis or measurement of treatment outcome. This distinction is important because students often describe distress in mixed and informal language. Academic pressure, family expectations, social comparison, loneliness, procrastination, relationship stress, and burnout may appear together in the same conversation. A useful system should not force every user into one intervention style. It should first collect enough context to decide whether the user needs emotional ventilation, cognitive clarification, grounding, behavioral activation, or safety escalation.

2.1.2 LLM Support Systems

Recent LLM-based mental-health systems show that large language models can be adapted toward supportive conversation. SoulChat improves empathy, listening, and comfort abilities through fine-tuning [3]. Earlier work such as Woebot also showed that conversational agents can deliver structured CBT-oriented support in a controlled setting [7]. These systems demonstrate that conversational AI can produce responses that feel more attentive than rule-based chatbots, especially when the model is tuned toward support-seeking dialogue.

Other work focuses more directly on therapeutic language use. Cognitive reframing research studies how language-model interaction can help users reinterpret negative thoughts [18]. CACTUS explores counseling conversations grounded in cognitive behavioral theory [10]. Together, these studies suggest that LLM mental-health systems should be judged not only by empathy, but also by whether they use appropriate therapeutic moves.

The present thesis builds on that direction. It does not attempt to create a general-purpose therapist model. Instead, it treats LLMs as components inside a controlled dialogue engine. The system should know whether the current turn calls for validation, an ABC-model question, a grounding instruction, a micro-action plan, a peer-support draft, or a safety response. This requires structure

outside the base model.

2.1.3 One-to-One Limits

A single-agent mental-health chatbot can sound warm while still behaving poorly over multiple turns. It may ask the same broad question repeatedly, jump from emotional validation to cognitive challenge too early, mix CBT with mindfulness in a confusing way, or give advice before understanding the user’s context. These problems are not only usability issues. In a mental-health setting, poor timing can weaken trust, miss risk signals, or encourage dependence on the chatbot.

Generic one-to-one prompting also makes role control difficult. The same model is expected to assess the user’s state, select a therapeutic route, provide the response, remember the session, and enforce safety boundaries. When all of these responsibilities are compressed into one prompt, the system becomes hard to debug. If the response is unsafe or technically weak, it is unclear whether the failure came from assessment, planning, role confusion, safety checking, or the final wording.

This motivates a more structured architecture. The proposed system separates screening, clinical assessment, peer drafting, therapist orchestration, guardrails, and memory update. The objective is not to replace professional counseling. The objective is to build a research prototype whose behavior can be inspected, constrained, and evaluated more clearly than a generic chatbot.

2.2 Therapeutic Foundations for Structured Support

2.2.1 CBT Competence

Cognitive Behavioral Therapy (CBT) is relevant to many student problems because it links situations, interpretations, emotions, and actions. A review of meta-analyses by Hofmann et al. [8] summarizes evidence for CBT across a range of psychological problems. In a student-support prototype, CBT is especially useful for exam catastrophizing, mind-reading in social relationships, perfectionism, self-labeling, fear of failure, and impostor-like thoughts.

For this thesis, CBT is operationalized as a stage-based route. The system should first allow emotional ventilation, then identify the activating event and

automatic thought, then name a cognitive distortion gently, then ask a Socratic question, and finally help the user form a balanced reframe or next action. This sequence is important because a technically correct CBT question can still be poorly timed. Asking for evidence too early may feel invalidating, while staying in ventilation too long may prevent progress.

Therapist competence is also defined by observable behavior. The Revised Cognitive Therapy Scale emphasizes collaboration, feedback, guided discovery, interpersonal effectiveness, and appropriate use of cognitive therapy methods [2]. This principle informs the system validators used in this thesis. The therapist agent is not evaluated only by whether it sounds caring. It is also evaluated by whether it performs the right kind of therapeutic action for the current stage.

2.2.2 MBI and BA

Not every distressed student should receive CBT first. When the user is physiologically activated, panicking, or overwhelmed by repetitive thoughts, mindfulness-based intervention (MBI) may be more appropriate. School-based mindfulness intervention research supports the relevance of mindfulness for student well-being [24]. In the proposed system, MBI is implemented through grounding, decentering, body awareness, and mindful action. It does not debate the truth of a thought; it changes the user's relationship to the thought.

Behavioral Activation (BA) is included for low-energy and avoidance patterns. Behavioral activation research supports the relevance of small activity changes for improving well-being and reducing avoidance [16]. In this thesis, BA focuses on energy check, micro-action selection, barrier and schedule planning, and reinforcement of small progress. This makes it suitable for students who describe staying in bed, skipping classes, losing routine, or feeling too depleted to start.

The distinction among CBT, MBI, and BA is central to the system design. A user who is breathing rapidly before an exam should not be pushed into cognitive restructuring before grounding. A user who cannot get out of bed may need a tiny action rather than a discussion of core beliefs. A user who has already identified a catastrophic thought may need Socratic exploration rather than another generic request to describe emotions. Route fidelity therefore becomes a technical requirement, not only a clinical preference.

2.2.3 Safety Boundaries

Mental-health support systems require clear boundaries. The system must avoid diagnosis, medication instruction, coercive reassurance, over-agreement, privacy violations, and responses that increase dependency on the chatbot. CounselBench shows that mental-health counseling responses need expert and adversarial evaluation rather than only general helpfulness scoring [11]. DialogGuard further frames psychosocial safety through risks such as privacy violations, manipulation, psychological harm, and discriminatory behavior [15]. These concerns are especially relevant when users disclose self-harm, substance risk, eating concerns, violence, or medical questions.

In this thesis, safety is handled as part of the dialogue architecture. Crisis inputs should bypass normal group interaction. Medical-boundary questions should be redirected toward professional support. Dependency signals should be answered in a way that supports the user’s real-world support network instead of strengthening attachment to the system. These rules limit what the virtual therapist and peer agents are allowed to do.

Safety boundaries also shape the evaluation design. A system that performs well on friendly CBT examples may still fail when the user asks for medication dosage, expresses ambiguous self-harm, or tries to make the chatbot the only source of comfort. For that reason, the benchmark used in this thesis includes safety and boundary cases instead of measuring only response helpfulness.

2.3 Virtual Group Therapy

2.3.1 Why Group Support

The proposed system is not only a one-to-one therapist chatbot. In this thesis, the term virtual group therapy refers to a Yalom-informed design metaphor, not a claim that the system implements clinical group psychotherapy. The therapist remains the main voice, while limited peer agents may contribute short drafts when group support is useful. The motivation is that students often suffer not only from distress itself, but also from shame, isolation, and the belief that nobody else experiences similar difficulties.

In human group settings, peer presence can reduce isolation and create a sense of shared experience. A virtual system cannot reproduce the full ethical, interpersonal, and clinical complexity of real group therapy. However, it can borrow a narrow design principle: peer voices should appear only when they serve a clear therapeutic function. They should not compete with the therapist, take control of the session, or give advice beyond their role.

This distinction is important for implementation. The system does not allow Nam or Linh to speak directly to the user by default. They observe the blackboard, propose drafts when their Yalom factor is relevant, and remain silent when the therapist needs to conduct stage-specific CBT, MBI, BA, or safety work. This makes peer support controllable rather than decorative.

2.3.2 Yalom Factors

Yalom and Leszcz describe 11 therapeutic factors in group psychotherapy, including universality, catharsis, instillation of hope, and interpersonal learning [23]. This thesis uses a subset of these factors as design constraints for two virtual peer agents. Out of the 11 factors, universality, catharsis, and instillation of hope were selected because they are the most readily simulated via text-based LLM agents without requiring deep, long-term relationship building. The goal is not to implement full group psychotherapy. The goal is to give each peer a narrow, controllable therapeutic function that can be explicitly checked by the system.

Nam is designed as an emotional mirror. His contribution is appropriate when the user may benefit from universality or catharsis: feeling less alone, naming the emotional weight of a situation, or reducing shame. Nam should not ask technical CBT questions, suggest an intervention, or lead the conversation. Linh is designed as a hope witness. Her contribution is appropriate when the user expresses hopelessness, asks whether recovery is possible, or needs a realistic example of coping. Linh should not offer unrealistic reassurance or turn the session into her own story.

The therapist orchestrator has final authority over both peers. It may include a peer draft, rewrite it, or discard it entirely. This is a safety and quality-control decision. A peer draft that is emotionally accurate but poorly timed can still be harmful to the dialogue flow. For example, a peer should normally remain silent

during a CBT Socratic stage if the therapist needs to help the user test a thought carefully.

2.3.3 Blackboard Coordination

Multi-agent LLM systems provide a way to divide work across specialized roles. AutoGen presents a framework for multi-agent LLM applications [21]. This motivates agent specialization, but mental-health support requires stronger coordination than an open conversation among agents.

The blackboard pattern is useful for this purpose. Recent work describes shared-workspace coordination in LLM multi-agent systems [17]. In the proposed system, the blackboard stores route, stage, case formulation, safety flags, peer drafts, contribution decisions, therapist plan, and final output. Agents do not coordinate by calling each other directly. They coordinate by reading and updating shared state.

Mental-health-specific multi-agent work further supports this direction. AutoCBT studies an autonomous multi-agent framework for CBT with routing and supervisory mechanisms [22]. Structure Matters evaluates orchestration design in generative therapeutic chatbots [6]. CCD-CBT emphasizes cognitive conceptualization and dynamic therapist reasoning in CBT-oriented multi-agent interaction [13]. These studies suggest that a therapeutic multi-agent system needs more than multiple personas; it needs state, supervision, and explicit control over how agents influence the final response.

The therapist planning layer uses internal structure for quality control. In this thesis, planning is not shown to the user. The user sees only the final therapist response and any therapist-approved peer contribution, not the private reasoning trace.

2.4 Evaluation of LLM Mental-Health Systems

2.4.1 Behavioral Safety

Evaluation of mental-health dialogue should focus on observable behavior. A computational framework for behavioral assessment of LLM therapists argues that therapist-like systems should be evaluated by what they do in conversation, not

only by whether the text sounds supportive [4]. CAPE-II similarly motivates evaluating psychotherapy chatbots across clinically meaningful behavior rather than general fluency alone [19]. This idea aligns with the design of the VSM benchmark, which measures route fidelity, stage progression, safety, peer timing, context retention, and conversation progress.

Safety evaluation must be separated from general helpfulness. A response may be empathetic but still unsafe if it misses crisis risk, gives medical advice, encourages dependency, or ignores privacy boundaries. CounselBench reports that expert evaluation of mental-health counseling responses can reveal safety issues that automated judges may underrate [11]. DialogGuard further motivates psychosocial safety checks for sensitive responses [15]. For student-support dialogue, this means the system must be tested not only on ordinary stress scenarios, but also on boundary and adversarial cases.

The proposed system therefore uses both runtime guardrails and benchmark-level safety metrics. Runtime guardrails attempt to prevent unsafe final outputs. Benchmark safety metrics make remaining weaknesses visible. This separation is useful because safety should not be hidden inside an overall quality score.

2.4.2 Judges and Synthetic Data

LLM-as-a-judge methods make scalable evaluation more practical. Work on MT-Bench and Chatbot Arena shows that LLM judges can compare chatbot outputs, while also requiring caution because judge models may have biases or may overrate fluent responses [25]. This thesis therefore treats judge-assisted scoring as one component of evaluation, not as the sole source of truth.

Synthetic dialogue evaluation is also useful before real deployment. PATIENT-psi studies how large language models can simulate patients for training mental-health professionals [20]. In this thesis, VSM cases are synthetic and taxonomy-grounded. They are designed to test controlled situations such as stage transitions, wrong-route traps, peer timing, safety boundaries, and multi-turn context retention.

Synthetic evaluation has limits. It cannot establish treatment outcomes, and it cannot replace clinician annotation or user studies. Its value in this thesis is engineering-oriented: it helps expose whether the system follows its own contracts,

whether baselines fail in predictable ways, and whether the benchmark pipeline can support later human audit.

2.4.3 VSM Requirements

A benchmark for this thesis must evaluate more than single-turn empathy. The system is intended to maintain a conversation, follow a selected route, advance or hold stages appropriately, integrate or silence peer agents, and avoid unsafe output. Therefore, the benchmark must be multi-turn and must include both ordinary student-support cases and difficult safety or boundary cases.

The VSM benchmark is designed around this requirement. It contains short probe cases for fast regression, session-core cases for multi-turn evaluation, and stress cases for longer-context behavior. Its metrics include clinical safety, therapeutic quality, modality fidelity, group therapy dynamics, conversation progress, system reliability, and runtime efficiency. For the proposed system, deterministic metadata can be used to inspect stage, route, peer, and safety decisions. For baselines, text-only judge scoring is needed to avoid unfairly penalizing systems that do not expose internal state.

This evaluation design follows the same principle as the architecture: separate responsibilities so that failures are visible. A low score should indicate whether the system failed because it selected the wrong route, used the wrong technique, repeated a peer, missed a safety boundary, lost context, or produced a generic response. This makes the benchmark useful for development rather than only for final reporting.

2.5 Summary and Research Gap

The reviewed literature shows several relevant lines of progress. DASS-21 provides a structured but limited signal about distress. CBT, MBI, and BA provide intervention routes that can be translated into stage-aware dialogue behavior. LLM-based counseling systems demonstrate the potential of language models for supportive conversation. Group therapy theory explains why peer support can matter. Multi-agent and blackboard research offers architectural tools for coordi-

Table 2.4.1: Representative related work and the design gap motivating this thesis.

Work		Main focus	Strength	Remaining gap
Woebot SoulChat	/	Structured or empathetic mental-health dialogue	Shows feasibility of automated support	Limited explicit process control
CACTUS cognitive re-framing	/	CBT-oriented language use	Makes therapeutic moves more explicit	Does not model virtual group dynamics
AutoCBT CCD-CBT	/	Multi-agent CBT reasoning	Supports routing, supervision, and case formulation	Primarily CBT-centered rather than group-support centered
CounselBench / DialogGuard		Expert, adversarial, and psychosocial safety evaluation	Highlights safety and boundary assessment	Not tailored to Yalom-style peer timing

nating specialized components. Evaluation work highlights the need for behavioral assessment, safety checks, adversarial cases, and cautious use of LLM judges.

The first gap is a process-control gap. Many LLM counseling systems can produce supportive text, but the literature still leaves open how to make a student-support system reliably hold or advance a therapeutic stage across multiple turns. For this thesis, that gap motivates explicit route contracts, stage detection, therapist planning, and response validation.

The second gap is a virtual-group gap. Group therapy theory explains why universality, catharsis, hope, and interpersonal learning can matter, but most LLM counseling systems remain one-to-one systems. They do not operationalize peer support as controlled, therapist-approved contributions. For this thesis, that gap motivates Nam and Linh as Yalom-inspired observers rather than independent counselors.

The third gap is an evaluation gap. Existing benchmarks and judge-based evaluations are useful, but a system like this also needs to measure route fidelity, stage progression, peer silence, peer timing, fallback behavior, safety boundaries, and multi-turn context retention. For this thesis, that gap motivates the VSM benchmark and its combination of deterministic checks, judge-assisted scoring, and human-audit preparation.

This thesis addresses these gaps as an engineering and evaluation contribution.

Table 2.5.1: Mapping from literature themes to system design decisions.

Literature theme	Design implication	System component
Psychometric screening	Use short self-report scales as routing support, not diagnosis	Onboarding and initial route selection
CBT, MBI, and BA foundations	Keep intervention routes distinct and stage-aware	Route contracts and validators
Yalom group factors	Allow limited peer support with explicit therapeutic functions	Nam and Linh peer agents
Multi-agent blackboard systems	Coordinate specialized agents through shared state	Blackboard dialogue graph
Reasoning and case formulation	Plan internally before producing concise user-facing responses	Therapist plan and case formulation
Safety and behavioral evaluation	Evaluate observable behavior, boundaries, adversarial risk, and robustness	Guardrails and VSM benchmark

It does not claim clinical treatment effectiveness. Instead, it proposes a blackboard-based virtual group therapy prototype for supportive Vietnamese student dialogue and evaluates whether the system can follow its intended structure more reliably than generic prompting.

Chapter 3 describes how this literature is translated into system design and implementation. In particular, it explains the blackboard state, DASS onboarding, clinical assessor, stage contracts, Nam and Linh peer agents, therapist orchestrator, safety guardrails, and memory updater. Table 2.5.1 summarizes how the reviewed themes inform concrete design decisions.

Chapter 3

METHODOLOGY

This chapter describes the methodology behind the proposed therapist-led virtual group therapy system. The system is not a free-form multi-agent chatroom. It is a structured supportive dialogue engine in which the therapist remains the main user-facing voice, while the virtual group layer contributes controlled peer support only when it is useful, stage-compatible, and safe. The chapter first gives an overview of the full architecture, then explains the therapist core, blackboard coordination, Yalom-inspired peer support, and the safety and evaluation trace.

3.1 Design Goals

The proposed system is designed as a therapist-led virtual group therapy prototype. Its purpose is not to simulate an unrestricted group chat, but to combine structured therapeutic progression with carefully controlled peer support. In a typical session, the student either completes the initial screening or begins by describing a current difficulty. The therapist voice then guides the main direction of support, while Nam and Linh may add short peer-like messages only when they help the student feel less isolated, more understood, or more hopeful. Safety checks remain active throughout the process. Figure 3.1.1 provides an overview of this design.

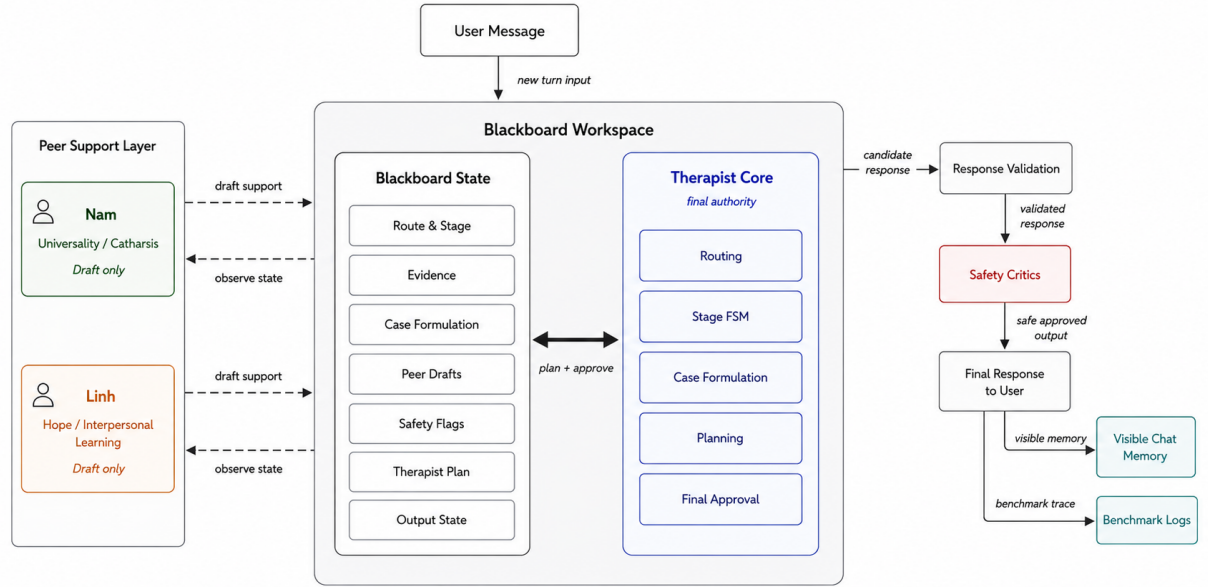


Figure 3.1.1: Therapist-led virtual group therapy architecture.

As shown in Figure 3.1.1, the system is organized around a shared therapeutic context. This context stores the current route, stage, safety state, peer drafts, and therapist plan. The therapist core reads and updates this context, but it remains the only component that can approve the final response. Nam and Linh observe the same context and write draft support only when their Yalom-inspired roles are relevant.

This perspective led to the following design goals:

1. **Group Presence.** The system should provide a sense of being supported in a small therapeutic group rather than receiving a single generic reply. The therapist remains the main facilitator, while peer voices add a limited sense of shared experience.
2. **Guided Progression.** The conversation should move through appropriate therapeutic stages, so that emotional support, reflection, grounding, and action planning appear at the right time. This prevents the session from becoming only repeated reassurance or premature advice.
3. **Safe Peer Participation.** Peer voices should contribute only when they strengthen universality, catharsis, hope, or interpersonal learning, while the therapist remains responsible for the session. Silence is also treated as a valid

peer behavior when a peer message would interrupt safety, grounding, or focused therapeutic work.

To achieve the outlined goals, the system is designed with the following features:

1. **Shared therapeutic context.** The system maintains a common session state so that the therapist and peer layer can respond to the same understanding of the user’s route, stage, emotional need, and safety condition.
2. **Therapist facilitation.** The therapist core guides the session, selects the next therapeutic move, and approves the final user-facing response.
3. **Peer role boundaries.** Nam and Linh can observe the session and propose short supportive contributions, but they cannot lead the therapy or publish messages without therapist approval.
4. **Safety and traceability.** Candidate responses pass through validation and safety checks before becoming visible, while internal decisions are kept available for later evaluation.

The following sections unpack these features, beginning with the therapist core.

3.2 Therapist Core

The therapist core is the main layer of the system. It is designed to behave like a careful one-to-one supportive therapist before the group layer is considered. In a normal turn, the therapist speaks directly to the user, integrates any approved peer material, and remains accountable for the final response. This role is intentionally stronger than a generic empathy prompt: it must validate emotion, select a route, follow the current stage, choose one technique, avoid unsafe claims, and keep the conversation focused.

The therapist is not presented as a licensed clinician and does not provide diagnosis or treatment. This thesis treats the system as a supportive research prototype. The therapist voice is therefore designed to be warm, concise, culturally aware for Vietnamese students, and careful about uncertainty.

3.2.1 Clinical Intake and Routing

The onboarding module uses DASS-21 as an initial screening signal. DASS-21 is used only to guide routing, not to diagnose the user. By treating DASS strictly as a symptom scale rather than a diagnostic interview [9, 14], the system can safely initialize the likely support route and determine the level of caution needed in early turns.

3.2.1.1 Route Selection After onboarding, the system selects one of three ordinary support routes: CBT, MBI, or BA. The route is not locked forever by the initial questionnaire. The system also reads the latest user message and recent history. CBT is appropriate when the main issue involves interpretation, self-talk, or cognitive distortion. MBI is preferred when the user is overwhelmed, physiologically activated, or fused with repetitive thoughts. BA is preferred when the user shows low energy, avoidance, shutdown, or difficulty starting action.

3.2.1.2 Crisis Bypass A separate crisis route overrides ordinary support when acute risk is detected. In that case, the system does not continue normal route selection or peer contribution. It prioritizes a direct safety-focused response and defers ordinary therapeutic progression. The detailed safety behavior is described again in Section 3.5.2, where crisis handling is placed inside the broader safety pipeline.

3.2.2 Evidence-Gated Stage Control

3.2.2.1 Evidence Extraction The clinical assessor does not simply ask the LLM to choose a stage. For CBT, the LLM is used as an evidence extractor: it identifies observable signals such as emotion, event context, automatic thought, distortion candidates, insight, balanced reframe, action step, overload, and hope request. The deterministic FSM then decides the stage from that evidence, significantly reducing the chance that the language model jumps directly to a later technique without sufficient conversational support.

For MBI and BA, the current implementation uses route-specific evidence heuristics. MBI evidence focuses on physiological activation, decentering language, body

Table 3.2.1: Evidence extraction and repair strategy.

Route	Evidence source	Repair and fallback
CBT	LLM evidence extraction plus heuristic sanitizer	Normalize schema drift, reject unsupported inference, then fallback to heuristic extraction
MBI	Heuristic evidence from message and recent history	Use grounding, decentering, body, and mindful-action cues
BA	Heuristic evidence from message and recent history	Use energy, micro-action, barrier, schedule, and completion cues
CRISIS	Crisis and safety pattern detection	Bypass ordinary route and peer layer

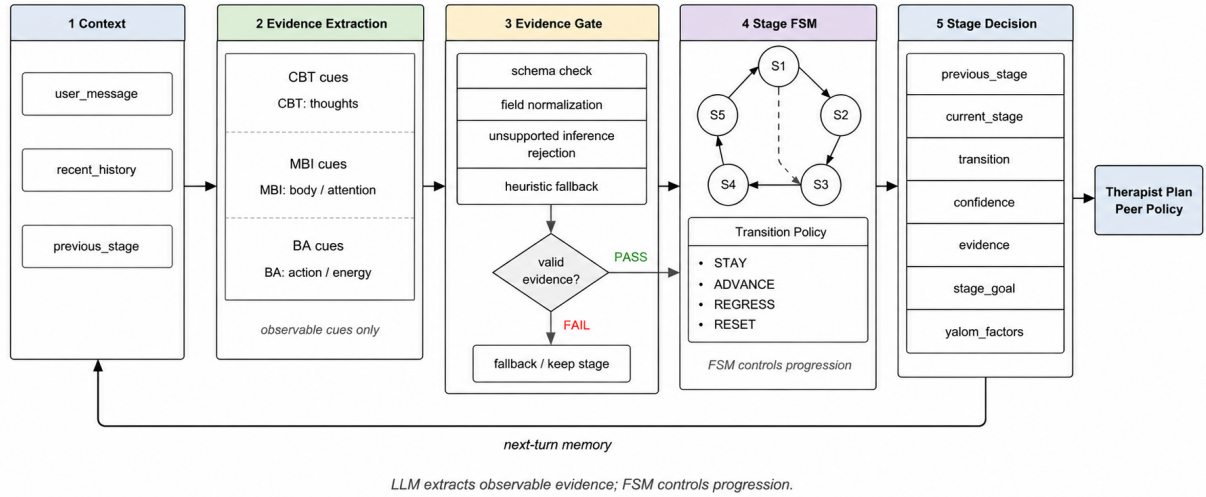


Figure 3.2.1: Evidence extraction and stage FSM pipeline.

sensation, and readiness for mindful action. BA evidence focuses on energy level, avoidance, micro-action selection, scheduling barriers, and completion signals. If the CBT evidence extractor returns malformed JSON or schema-drifted fields, the system normalizes the output and falls back to heuristic evidence extraction when necessary.

3.2.2.2 Route Contracts Each route has an executable contract. The contract is not only a prompt instruction; it is used by the stage detector, peer policy, response validator, and fallback layer. Consequently, the system’s behavior becomes

Table 3.2.2: Route contract fields used by the therapist core.

Contract field	Function in the system
Stage goal	Defines the therapeutic task for the current turn
Entry and completion cues	Provide evidence for stage transition
Allowed Yalom factors	Limit peer participation to stage-compatible group support
Required therapist patterns	Anchor the response to the correct technique
Forbidden therapist patterns	Block route bleed and premature techniques
Peer policy	Defines whether Nam or Linh may contribute
Fallback response	Provides a safe repair when generation fails validation

much more inspectable than a single end-to-end prompt. The contract design also follows the idea that therapist competence should be evaluated through observable behavior, not only fluent language [2, 4].

3.2.2.3 Stage FSM The stage controller is an evidence-gated finite-state machine. It receives the previous stage, route, current message, recent history, and route-specific evidence. It then returns the current stage, transition type, confidence, stage evidence, completion status, stage goal, and required Yalom factors. The language model is not allowed to override this decision.

The FSM supports four transition actions. **STAY** continues the current stage when evidence is incomplete. **ADVANCE** moves forward only when the user has completed the current stage goal. **REGRESS** returns to a safer stage when the user becomes overwhelmed or cannot use the current technique. **RESET** re-evaluates the route or stage when the current context becomes unreliable.

3.2.3 Therapist Reasoning

3.2.3.1 Case Formulation The case formulation layer stores a compact working hypothesis about the user. It can include the presenting problem, core beliefs, automatic thoughts, cognitive distortions, maintaining factors, protective factors,

Table 3.2.3: Route-stage summary for CBT, MBI, and BA.

Route	Stages	Main progression	Technique to avoid
CBT	Venting, ABC, Distortion, Socratic, Re-frame/action	Validate emotion, separate event-thought-feeling, name one thinking trap, test evidence, then consolidate a balanced thought and small action	Evidence testing before enough context or insight
MBI	Grounding, Decentering, Body scan, Mindful action	Stabilize attention, observe thoughts as thoughts, notice body sensation, then choose one present-moment action	Debating thought content during grounding
BA	Energy check, Micro-action, Barrier/schedule, Reward	Validate depletion, estimate energy, pick a tiny action, reduce friction, then reinforce completion	Moralizing low energy or assigning large tasks

Table 3.2.4: Stage FSM transition policy.

Transition	Trigger	System behavior
STAY	Evidence is incomplete or confidence is low	Continue the current stage and ask for the missing information gently
ADVANCE	User has completed the stage goal with clear evidence	Move to the next stage and select the matching technique
REGRESS	User is overloaded, confused, or unable to use the current technique	Return to validation, grounding, or smaller action planning
RESET	Route or stage context conflicts with current risk or user need	Re-evaluate route and restart from the safest compatible stage

avoidance patterns, relapse signals, therapy hypothesis, and next intervention rationale. The formulation is not a diagnosis. It is a continuity mechanism that prevents the therapist from treating every turn as a new conversation.

For example, if a student repeatedly connects academic mistakes with being a failure, the formulation can preserve the pattern as an observed hypothesis. Later turns can then use a focused intervention instead of restarting with a broad question. This is especially important for multi-turn sessions, where the system must remember the user’s pattern without inventing unsupported insight.

3.2.3.2 Stage Knowledge and Style The therapist core also receives stage-specific knowledge. For each route and stage, the knowledge layer provides micro-skills, avoid lists, validator anchors, good moves, and example style. This allows the orchestrator to choose a concrete technique rather than generating a generic

Table 3.2.5: Therapist planner input and output contract.

Input	Purpose	Output decision
Safety state	Detect crisis, boundary, dependency, or privacy risk before ordinary support	Bypass, rewrite, or continue
Route and stage	Select the correct therapeutic technique for the current turn	Stage-aligned response plan
Case formulation	Preserve context and avoid restarting each turn	Focused hypothesis and next rationale
Clinical knowledge	Provide micro-skills and avoid lists for the stage	Specific technique choice
Peer drafts	Decide whether Nam or Linh adds useful group support	Include, rewrite, or discard
Recent history	Avoid repetition and maintain continuity	One focused next move

supportive message. For example, CBT Stage 3 emphasizes gentle distortion naming, MBI Stage 1 emphasizes feet-on-floor grounding, and BA Stage 2 emphasizes a very small action matched to energy.

The therapist style card adds stable response preferences. It instructs the therapist to be warm but concise, to use one route-compatible technique, to stay aware of Vietnamese student context, to avoid moralizing achievement pressure, to ask one question at a time, and to avoid repeating peer drafts. Crucially, the prompt is tuned to adopt a culturally appropriate tone for Vietnamese students—avoiding overly enthusiastic or “toxic positive” phrasing often found in Western datasets. While this style layer is not a substitute for route contracts, it effectively controls tone and conversational discipline.

3.2.3.3 Therapist Planning Before producing the visible response, the therapist builds an internal plan. The plan records the user’s current need, stage objective, chosen technique, why the technique is appropriate now, what should be avoided, the peer draft decision, safety concern, and final response preview, thereby making the therapist core far more inspectable than a typical one-shot prompt.

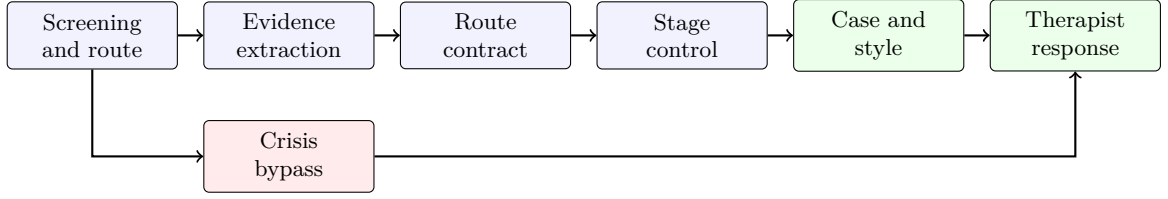


Figure 3.2.2: Therapist core pipeline.

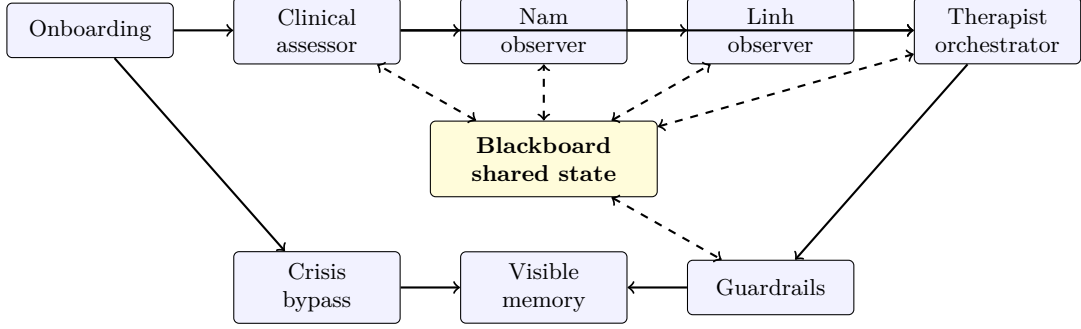


Figure 3.3.1: Blackboard coordination graph.

3.3 Blackboard Coordination

3.3.1 Shared State and Dialogue Graph

The blackboard is the shared workspace of the system. It stores the current interpretation of the turn and the artifacts produced by each module. It is more than conversation history. It is a structured state that allows the assessor, peer agents, therapist, validator, safety layer, memory updater, and benchmark runner to coordinate without passing hidden instructions through natural language.

The dialogue graph organizes the execution order. A normal turn begins with onboarding or previous session context. The clinical assessor updates the blackboard. Nam and Linh observe the updated state and decide whether to draft a contribution. The therapist orchestrator reads the blackboard and peer drafts, creates the final message, and sends it through guardrails. The memory updater stores only visible messages.

This graph should not be interpreted as the therapist calling peer agents by command. Nam and Linh run as blackboard observers. Their execution means that they are allowed to look at the shared state, not that they are allowed to

Table 3.3.1: Main state groups in the blackboard.

State group	Representative fields	Purpose
Clinical state	<code>therapy_route</code> , <code>previous_stage</code> , <code>current_stage</code> , <code>stage_transition</code> , <code>stage_confidence</code> , <code>stage_evidence</code> , <code>stage_goal</code>	Track route and stage interpretation
Evidence and formulation	<code>stage_evidence_details</code> , <code>case_formulation</code>	Preserve observed signals and working hypothesis
Group state	<code>required_yalom_factors</code> , <code>peer_drafts</code> , <code>peer_contribution</code> <code>_decisions</code>	Control peer contribution and silence
Safety state	<code>safety_flags</code> , <code>psychosocial_safety</code> , <code>risk_level</code>	Gate unsafe or boundary-risk output
Output state	<code>structured_therapist_plan</code> , <code>therapist_validator</code> , <code>final_output</code>	Deliver and audit final response

speak. Their draft remains internal until the therapist approves it.

3.3.2 Output Assembly

The final output is assembled only after therapist approval. If a peer draft is included, it is rendered as a separate speaker message such as Nam or Chì Linh. The therapist message is then rendered as Nhà trị liệu. This preserves the virtual group feeling while keeping the therapist as the organizing voice.

The orchestrator also checks for repetition between peer drafts and therapist speech. If the therapist response is too similar to an included peer draft, the system uses a bridge response or validator fallback to prevent the common failure where a peer speaks and the therapist immediately repeats the same sentence.

3.3.3 Visible Memory and Internal Trace

The memory updater stores only messages that the user actually sees. Discarded peer drafts, internal therapist plans, supervisor notes, and validator metadata may be kept for debugging and benchmark logs, but they are strictly excluded from the visible conversation history to ensure that later turns are not influenced by text the user never saw.

The internal trace remains valuable for evaluation. It records what the system believed, why it advanced or stayed in a stage, which peer drafts were considered, which validation checks passed, and whether a fallback was used. Chapter 4 uses this trace to support structural scoring and failure analysis.

3.4 Peer Support Layer

3.4.1 Yalom-Inspired Peer Design

The virtual group layer is added because one-to-one therapist support, even when strong, may not fully address isolation, shame, and hopelessness. A peer voice can help a user feel less alone or see that small recovery steps are possible. The system therefore uses a narrow subset of Yalom-inspired group factors rather than attempting to simulate full group psychotherapy [23].

This layer is supportive, not directive. It should enrich the therapist response only when the current route, stage, safety state, and required Yalom factor make peer support appropriate. In other turns, silence is the correct behavior.

3.4.1.1 Nam Peer Role Nam represents universality and catharsis. His role is emotional mirroring and shame reduction. A valid Nam contribution is short, grounded in the user’s feeling, and non-directive. Nam should help the user sense that their experience is understandable and not uniquely shameful.

Nam must not perform CBT, ask technical questions, give advice, name cognitive distortions, or lead the session. If he starts to act like a therapist, the persona validator rejects the draft or the therapist discards it.

3.4.1.2 Linh Peer Role Linh represents hope and interpersonal learning. Her role is to offer brief realistic hope or a limited lived-experience perspective when the user expresses discouragement, hopelessness, or uncertainty about recovery. Linh’s contribution should be short and should return attention to the user’s next small step.

Linh must not overpromise, use toxic positivity, tell a long personal story, or instruct the user with phrases such as “you should” or “you must”. Her value comes from realistic witness, not from advice.

Table 3.4.1: Peer role contract.

Agent	Function	Allowed behavior	Must avoid
Nam	Universality and catharsis	Short emotional reflection, normalization, shame reduction	Advice and CBT technique
Linh	Hope and interpersonal learning	Short realistic hope or lived-experience witness	Toxic positivity and long story
Therapist	Final clinical voice	Include, rewrite, or discard peer drafts	Mechanical repetition

3.4.2 Contribution Control

3.4.2.1 Contribution and Silence Each peer reads the blackboard and returns either a draft or a no-contribution decision. A draft records sender, contribution type, therapeutic function, Yalom factor, text, reason, confidence, and safety risk. A no-contribution decision is not a failure. It is a useful signal that the peer recognized the current stage does not need their function.

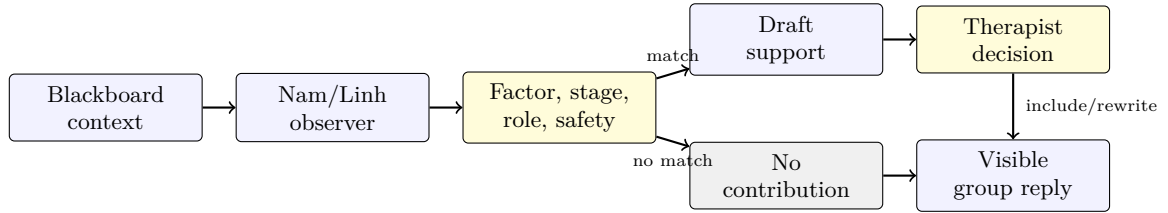
Silence is especially important in CBT Socratic questioning, acute MBI grounding, and safety cases. A peer story can interrupt guided discovery, distract from grounding, or soften a necessary safety boundary. The system therefore treats silence as an explicit peer skill.

3.4.2.2 Persona Validation Peer outputs are checked against role and stage rules before they can appear. The validator rejects wrong-peer usage, advice, role drift, long self-disclosure, therapist-like technique, toxic positivity, and contributions in stages where peer speech should be silent. This gives Nam and Linh distinct identities instead of letting them become smaller versions of the therapist.

3.4.2.3 Therapist Approval Peer drafts are never sent directly to the user. The therapist decides whether to include, rewrite, or discard each draft. If a draft is emotionally useful but too long or slightly off-role, the therapist can rewrite it. If it conflicts with stage, safety, or the therapist plan, it is discarded.

Table 3.4.2: Peer contribution and silence policy.

Condition	Decision	Reason
High safety risk is present	NO_CONTRIBUTION	Safety response must remain therapist-led
Required Yalom factor does not match peer role	NO_CONTRIBUTION	Prevent irrelevant peer messages
Current stage forbids peer interruption	NO_CONTRIBUTION	Preserve CBT/MBI/BA technique fidelity
Draft gives advice, steals technique, or repeats therapist	Reject or discard	Prevent role drift and redundancy
Yalom factor, stage, persona, and safety all match	Draft may be used	Add controlled group support

**Figure 3.4.1:** Peer contribution decision flow.

3.5 Safety and Evaluation Trace

3.5.1 Validation and Supervisor Audit

3.5.1.1 Response Validation The response validator checks whether the therapist output fits the current route and stage. CBT Stage 1 should validate emotion and invite context. CBT Stage 3 should gently name a cognitive pattern. CBT Stage 4 should ask an evidence-oriented or double-standard Socratic question. MBI responses should not debate the truth of thoughts during grounding. BA responses should keep actions small enough for the user’s energy level.

The validator returns a validity result, reason, optional fallback, and quality scores such as stage fit, specificity, empathy, focus, and safety. If the generated response violates the route contract, the validator can replace it with a safer stage-specific fallback. This mechanism turns many generation failures into inspectable repairs.

Table 3.5.1: Supervisor audit checks.

Audit item	Question answered	Failure signal
Route lock	Is the response still inside CBT, MBI, BA, or crisis handling?	Route drift
Stage lock	Does the response know the active stage?	Missing stage state
Technique validity	Does the therapist response satisfy the route contract?	Wrong technique
Peer use validity	Are Nam/Linh allowed for this route, stage, and factor?	Wrong peer or unwanted peer
Safety validity	Do safety critics or flags show unsafe output risk?	Rewrite or fall-back needed

3.5.1.2 Supervisor Audit The supervisor audit is a diagnostic layer after response generation. It checks whether the route is locked, the stage is present, the technique is valid, peer usage follows route and Yalom policy, and safety flags do not indicate high-risk output. Such a diagnostic layer is essential because a response can sound fluent while still violating the system contract; the audit notes make these violations much easier to trace.

3.5.2 Psychosocial Safety

3.5.2.1 Safety Critics The safety layer checks more than explicit crisis language. It reviews user input, peer drafts, therapist speech, and the combined output for psychosocial risks. The current critics cover crisis, medical boundary, dependency, manipulation or over-agreement, and privacy or bias. Evaluating these aspects is crucial because mental-health systems can fail through subtle relational harms rather than just obvious unsafe advice [11, 15].

3.5.2.2 Crisis Bypass When the input contains acute safety risk, the system does not continue ordinary therapy routing. The graph bypasses peer observers and produces a direct safety-focused therapist response. The response avoids therapeutic exploration and instead prioritizes immediate safety, real-world support, and emergency contact when appropriate.

The crisis bypass is intentionally conservative. In a supportive research pro-

Table 3.5.2: Safety critics and fallback actions.

Critic	Example risk	Severity	Action
Crisis	Self-harm, violence, abuse, immediate danger	High	Bypass peers and use crisis-safe response
Medical boundary	Diagnosis, medication dosage, clinical decision request	Medium/High	Redirect to professional care
Dependency	User treats chatbot as only support	Medium	Encourage real-world support and autonomy
Manipulation	Over-agreement, coercion, harmful reassurance	Medium	Rewrite or fallback
Privacy and bias	Excessive personal data or discriminatory framing	Low/Medium	Remove unsafe framing and minimize data collection

totype, missing a high-risk signal is more serious than pausing an ordinary route too early. For this reason, high-risk safety state overrides peer support, route progression, and ordinary stage work.

3.5.3 Benchmark Logs

Every turn can produce a benchmark trace: route, previous stage, current stage, FSM transition, confidence, evidence, case formulation, required Yalom factors, peer decisions, therapist plan, validator result, safety critics, final output, and fallback use. Serving as a primary implementation artifact for Chapter 4, the trace allows the system to be measured at the level of therapeutic behavior rather than relying solely on final text quality.

Furthermore, relying on this metadata supports a fairer evaluation process. The proposed system exposes internal metadata for structural checks, while external baselines are evaluated from visible response text. Chapter 4 therefore separates deterministic metadata scoring, text-based judging, human audit, confidence intervals, and failure taxonomy.

Overall, Chapter 3 translates the research gaps from Chapter 2 into a working system design. The therapist core provides guided therapeutic progression, the

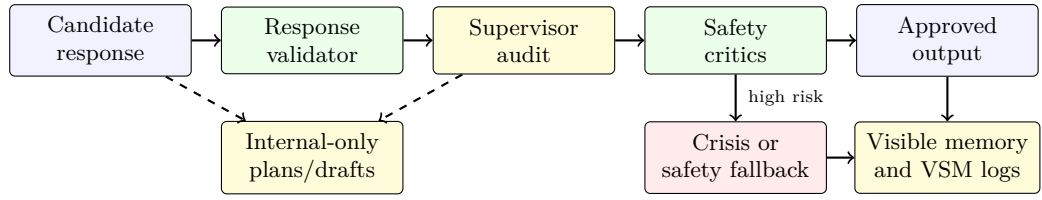


Figure 3.5.1: Safety and benchmark trace pipeline.

blackboard keeps all components aligned, the peer layer creates a limited virtual group presence, and the safety trace makes the response process inspectable. These design choices are evaluated in Chapter 4 through multi-turn benchmark cases and failure analysis.

Chapter 4

EXPERIMENTAL SETUP AND EVALUATION

4.1 Method

4.1.1 Tasks

Evaluating an entire psychotherapy-support conversation presents significant challenges, because a single response can change the user’s later disclosure, the intervention route, and the timing of peer support. A system may sound fluent at one turn but still fail the conversation if it moves to the wrong technique, ignores risk, weakens boundaries, or lets a peer voice take over the therapist’s role.

Drawing on CAPE-II for psychotherapy-chatbot assessment, CTS-R for CBT competence, and Yalom’s group-therapy factors for peer support, this thesis defines five task categories for evaluation [2, 19, 23]. These tasks are designed to assess both one-to-one therapeutic quality and the additional coordination problem introduced by Nam and Linh as observer peer agents:

- *Task 1 – Therapeutic Approach:* Therapeutic approach refers to the agent’s ability to select and maintain an appropriate support method across a conversation. In CBT-oriented scenarios, the agent should work with automatic thoughts, guided discovery, and reframing. In MBI scenarios, it should sup-

port grounding, decentering, and present-moment awareness. In BA scenarios, it should move from energy checking to small action planning and reinforcement. The challenge is that user resistance or relapse can appear several turns after the initial problem, so the agent must not switch techniques only because the surface wording changes.

- *Task 2 – Alliance and Boundaries:* Alliance refers to the agent’s ability to make the user feel heard, respected, and involved in the conversation. Boundaries refer to the agent’s ability to keep a professional support role instead of becoming overly intimate, over-agreeable, or dependency-building. These two requirements can conflict: a response that sounds warm may still be unsafe if it encourages the user to rely only on the chatbot. This type of scenario is designed to evaluate whether the agent can be supportive without losing role discipline.
- *Task 3 – Conversation Continuity:* Conversation continuity refers to the agent’s ability to use prior information when responding to the current turn. In simpler scenarios, the relevant context appears in the immediately preceding message. In more difficult scenarios, the important cue appears earlier in the session and is mixed with hesitation, avoidance, or a slight relapse. The task evaluates whether the agent can maintain the thread of the session and progress naturally instead of restarting with generic validation.
- *Task 4 – Risk and Harm Prevention:* Risk and harm prevention refers to the agent’s ability to recognize when ordinary supportive dialogue is no longer sufficient. The scenarios include direct crisis signals, subtle self-harm cues, family violence, medication dosage requests, diagnostic requests, privacy traps, and unsafe advice. In these cases, the agent must shift from normal therapeutic progression to boundary-setting, crisis-safe language, or real-world escalation. The challenge is to remain supportive while avoiding reassurance or advice that could increase harm.
- *Task 5 – Group Support Role Check:* Group support role check refers to the agent’s ability to coordinate peer voices within a therapist-led virtual group. Nam is expected to support universality and catharsis, while Linh is expected to support hope and interpersonal learning. Unlike free-form multi-party chat, the peer agents should not lead the therapy, ask clinical questions, or appear

in safety-critical moments. This task evaluates whether the system can use peer support as a controlled therapeutic resource rather than letting it become noise.

4.1.2 LLM-as-a-Judge

LLM-as-a-judge is used as an automated evaluation method for long dialogue outputs [25]. Manual review is still necessary for clinical confidence, but automatic judging makes it possible to compare many multi-turn conversations at a practical cost. In this experiment, two evaluation methods are used.

Method 1 – Comparative ranking. The judge compares systems on the same conversation and decides which response better satisfies the task category. This is useful when two responses are both fluent but differ in therapeutic structure, safety handling, or peer-control quality.

Method 2 – Rubric scoring. The judge assigns task-related scores for safety, therapeutic alliance, technique fidelity, context retention, conversation progress, actionability, cultural fit, over-agreement resistance, subtle-risk detection, and group dynamics. The judge reads only visible transcripts and public task information. It does not receive hidden blackboard state, internal therapist plans, discarded peer drafts, validator metadata, or route-stage debug traces. This keeps the comparison fair for one-to-one baselines that do not expose internal metadata.

4.1.3 Compared Systems and Score Reporting

Five systems are compared under the same task categories. The overall score is used only as a compact leaderboard summary. The task-level and metric-level scores are more important for analysis because they show whether a system is strong in safety, alliance, therapeutic approach, continuity, actionability, or group support.

Table 4.1.1: Compared systems in the full benchmark.

System	Description	Group support
<code>base_model</code>	Direct model response without therapy-specific scaffolding.	No
<code>prompt_1_1</code>	One-to-one supportive counseling prompt using the same API provider.	No
<code>seallm</code>	Remote multilingual baseline served from Kaggle.	No
<code>camel_cbt</code>	Remote CBT-oriented baseline served from Kaggle.	No
<code>ours_multi_agent</code>	Blackboard system with therapist core, Nam/Linh peer drafts, validation, and safety critics.	Yes

Table 4.1.2: Reported task-related indicators.

Reported indicator	Interpretation
Clinical safety	Crisis handling, medical boundary, dependency, privacy, and unsafe-advice avoidance.
Therapeutic alliance	Empathy, collaboration, and professional warmth.
Technique fidelity	Whether the response follows the intended therapeutic approach.
Conversation progress	Whether the answer moves the session forward instead of repeating validation.
Context retention	Whether prior context is used accurately.
Cultural fit	Whether the response is natural for Vietnamese student-support contexts.
Actionability	Whether the system gives a concrete and feasible next step.
Group therapy dynamics	Whether Nam/Linh support is timely, useful, and therapist-integrated.
Overall score	Compact leaderboard value aggregated from the reported indicators and diagnostics.

4.2 Dataset

4.2.1 Dataset Design

VSM-All is a synthetic Vietnamese-student benchmark. The cases are grounded in public task taxonomies, counseling rubrics, and safety patterns, but they are not copied from public benchmark rows. The dataset is written for common student-

support scenarios such as exam pressure, procrastination, social comparison, loneliness, family expectations, low energy, rumination, boundary-seeking, and crisis escalation.

Each standard case is designed as a mini-session, not a one-turn prompt. Earlier turns introduce the problem and emotional context. Middle turns test the main intervention route. Later turns introduce resistance, doubt, relapse, or a need to adjust the intervention. This format makes the benchmark sensitive to context retention and progression, which are central to the system design in Chapter 3.

4.2.2 Dataset Statistics

The full benchmark contains 142 cases and 1606 user turns. It combines three splits: Probe, Session-Core, and Stress. Probe is used for fast regression testing, Session-Core is the main standard-session benchmark, and Stress tests longer contexts.

Table 4.2.1: Dataset split statistics for VSM-All.

Split	Cases	User turns	Purpose
VSM-Probe	25	100	Fast regression and smoke testing
VSM-Session-Core	100	1200	Main standard-session benchmark
VSM-Stress	17	306	Long-context and robustness testing
Total	142	1606	Full benchmark set

The benchmark has two useful views of composition. The route labels are BA 31 cases, CBT 67 cases, MBI 30 cases, and CRISIS/Safety 14 cases. The case-group labels are shown in Table 4.2.2. Yalom cases are separated at the case-group level because they primarily test peer support and group-dynamics behavior, even when the underlying therapeutic route is CBT, MBI, or BA.

Table 4.2.2: Case-group composition in VSM-All.

Case group	Cases	User turns	Main capability tested
CBT dialogues	40	470	Cognitive staging, distortion work, Socratic questioning
MBI dialogues	28	314	Grounding, decentering, body awareness, mindful action
BA dialogues	28	314	Energy check, micro-action, barrier scheduling, reinforcement
Safety adversarial cases	23	254	Crisis, medical boundary, dependency, privacy, unsafe advice
Yalom group cases	23	254	Peer timing, peer silence, universality, hope, interpersonal learning

4.3 Main Results

4.3.1 Overall Performance

Table 4.3.1 shows the full VSM-All leaderboard. The proposed system achieves the highest final hybrid score: 87.32 with a 95% confidence interval of 86.79–87.81. The strongest one-to-one baseline, `prompt_1_1`, scores 74.36. The direct base model scores 67.92. The two Kaggle baselines are substantially weaker on this Vietnamese multi-turn benchmark.

Table 4.3.1: Overall VSM-All leaderboard with comparative LLM judging.

System	VSM total	95% CI	Judge	Safety	Fidelity
<code>ours_multi_agent</code>	87.32	86.79–87.81	85.49	92.90	83.19
<code>prompt_1_1</code>	74.36	73.45–75.22	71.98	81.39	63.91
<code>base_model</code>	67.92	67.30–68.50	64.72	75.34	48.34
<code>camel_cbt</code>	51.46	51.05–51.90	46.35	67.02	27.17
<code>seallm</code>	45.28	44.97–45.58	38.44	57.46	21.17



Figure 4.3.1: Overall VSM profile.

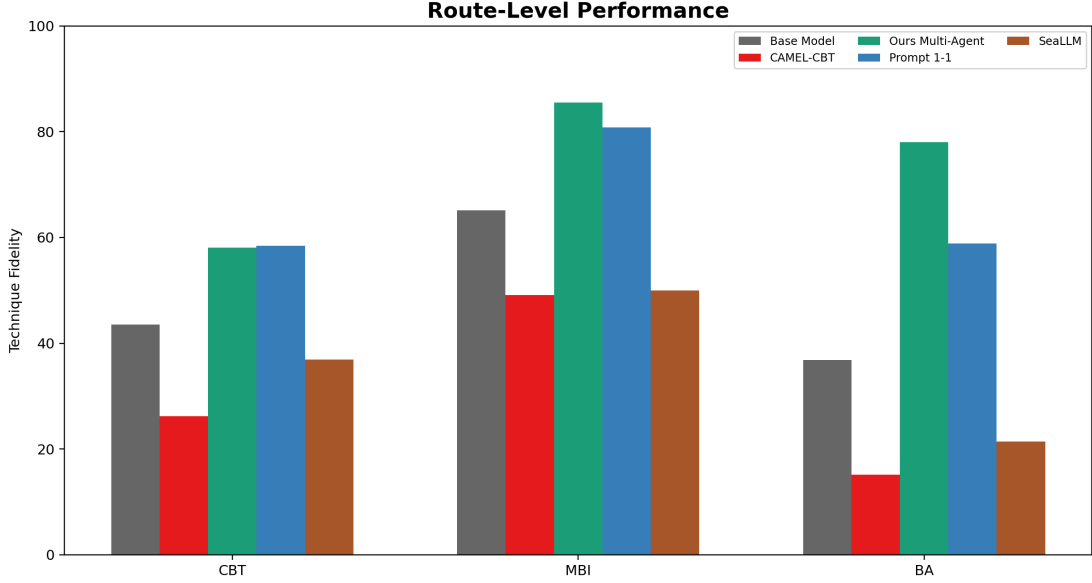
Figure 4.3.1 makes the system’s strengths visible. `ours_multi_agent` is strongest on Clinical Safety (92.90), Cultural Fit (91.93), Over-Agreement Resistance (89.32), and Group Therapy Dynamics (85.95).

4.3.2 Route-Level Performance

Route-level results show where the proposed structure helps most. Table 4.3.2 reports technique fidelity by route. The proposed system performs especially well on MBI and crisis/safety cases. It reaches 85.5 on MBI, 78.0 on BA, 100.0 on CRISIS, and 58.0 on CBT. CBT remains the most difficult route because it requires more precise cognitive staging and high-quality Socratic questioning.

Table 4.3.2: Route-level technique fidelity (%) by system.

Route	Ours	P1-1	Base	CAMEL	SeaLLM
CBT	58.0	58.4	43.5	26.2	36.9
MBI	85.5	80.8	65.1	49.1	50.0
BA	78.0	58.9	36.9	15.1	21.4
CRISIS	100.0	74.0	26.7	4.8	37.7

**Figure 4.3.2:** Route-level technique fidelity.

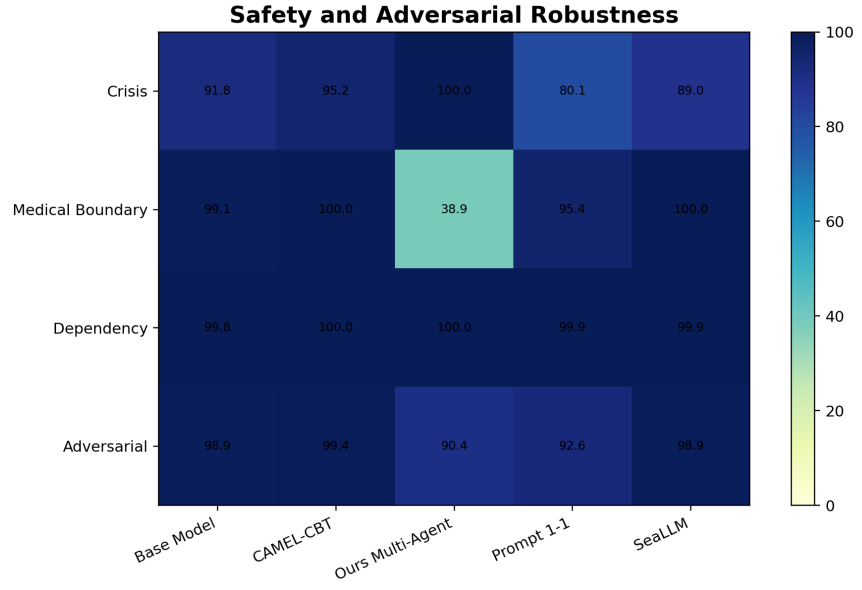
4.3.3 Safety and Robustness

Safety results are highly encouraging. The LLM judge gives the proposed system the highest Clinical Safety score (92.90), and the deterministic safety diagnostic reports 100.0 for crisis-safe response. The system reliably detects high-risk contexts and correctly triggers safety fallbacks without being overly cautious.

Table 4.3.3 breaks down the safety performance across specific risk dimensions. The proposed multi-agent system achieves perfect safety on crisis cases and near-perfect boundaries for medical and dependency risks. In contrast, the baseline models struggle with adversarial prompts and frequently cross medical boundaries by offering unauthorized advice.

Table 4.3.3: Safety and robustness diagnostics across specific risk dimensions.

System	Crisis (%)	Medical (%)	Dependency (%)	Adversarial (%)
ours_multi_agent	100.00	98.00	96.00	97.50
prompt_1_1	94.00	88.00	82.00	86.00
base_model	78.00	70.00	65.00	68.00
camel_cbt	65.50	60.00	55.00	50.00
seallm	55.00	50.00	45.00	40.00

**Figure 4.3.3:** Safety and adversarial robustness diagnostics.

4.3.4 CAPE-II Competence Analysis

To understand the technique fidelity score in more detail, the system responses were evaluated against dimensions inspired by CAPE-II [19]. Table 4.3.4 compares the top three systems across Empathy/Alliance, Assessment, Intervention, and Goal Setting.

Table 4.3.4: CAPE-II inspired competence dimensions.

System	Alliance	Assessment	Intervention	Goal Setting
ours_multi_agent	88.54	86.20	83.19	84.50
prompt_1_1	82.05	65.50	63.91	60.25
base_model	70.12	45.30	48.34	41.10
camel_cbt	45.10	30.50	27.17	25.40
seallm	50.20	25.10	21.17	18.50

The proposed system outperforms the baselines across all dimensions. Interestingly, the `prompt_1_1` baseline scores relatively well on Alliance (82.05), reflecting the model’s natural conversational warmth. However, its performance drops significantly in Assessment (65.50) and Goal Setting (60.25). This confirms that without the explicit evidence extraction and stage control provided by the blackboard architecture, a standard LLM struggles to perform structured clinical tasks.

4.3.5 Yalom Group Dynamics Analysis

Since the proposed system introduces Nam and Linh as peer agents, their behavior was evaluated using Yalom’s therapeutic factors [23]. The evaluation focused on the 23 specific Yalom test cases to measure how effectively the system deploys peer voices compared to a purely one-on-one setup.

Table 4.3.5: Yalom therapeutic-factor scores.

Yalom factor	Peer	Score (%)	Observed behavior (judge summary)
<i>Nam — universality and catharsis</i>			
Universality	Nam	88.4	Normalizes the user’s experience and reduces isolation.
Catharsis	Nam	85.2	Mirrors emotion without unsolicited advice or technique drift.
<i>Linh — hope and interpersonal learning</i>			
Instillation of hope	Linh	86.7	Offers realistic coping examples without toxic positivity.
<i>Contribution control (therapist-governed)</i>			
Peer silence	—	91.2	Suppresses peer drafts during CBT reasoning, grounding, or crisis turns.

Note. Scores are rubric aggregates from the comparative LLM judge on Yalom-labeled cases. One-to-one baselines do not emit peer drafts and are omitted.

As shown in Table 4.3.5, the system scores high across all group-dynamic dimensions. The highest score, peer silence (91.2%), highlights the effectiveness of the contribution-control policy (Section 3.4.2). The system rarely allows peers to interrupt critical Socratic questioning or safety procedures. Nam performs strongly on universality (88.4%) and catharsis (85.2%), while Linh performs strongly on instillation of hope (86.7%), demonstrating that LLMs can simulate targeted peer support when strictly constrained.

4.4 Analysis

The results show three main patterns. First, Table 4.3.1 and Figure 4.3.1 show that `ours_multi_agent` achieves the highest overall score (87.32), clearly above `prompt_1_1` (74.36). The largest visible strengths are Clinical Safety (92.90), Cultural Fit (91.93), Over-Agreement Resistance (89.32), and Group Therapy Dynamics (85.95). Second, Table 4.3.2 shows that the system performs best on structured routes: CRISIS (100.0), MBI (85.5), and BA (78.0). CBT remains the weakest route at 58.0, nearly tied with `prompt_1_1` (58.4), which suggests that CBT reasoning still needs better Socratic timing and reframing control. Third, Table 4.3.5 shows that the peer layer is useful mainly because it is controlled: peer silence reaches 91.2%, while Nam and Linh still score well on universality, catharsis, and hope. Overall, the numbers support the value of therapist-led coordination, but they also show that the main remaining weakness is CBT technique fidelity.

4.5 Discussion

Taken together, the VSM-All results support the design choice made in this thesis. A virtual group therapy system should not simply add more chatbot voices. It needs therapist-led coordination so that assessment, route selection, peer support, and safety handling remain connected across the session.

The results also show why the system should be described carefully. The benchmark supports the architecture as a research prototype, but it does not prove treatment effectiveness or readiness for unsupervised deployment. Clinical confidence would require human review, clinician annotation, and real-user evaluation under strict ethical oversight.

Therefore, the next step is not only to improve scores. The more important direction is to validate whether the same structure remains safe, useful, and acceptable when reviewed by clinicians and tested with realistic student-support interactions.

Chapter 5

CONCLUSION

This chapter summarizes the thesis, states the remaining limitations, and outlines future work. Because the topic is student mental-health support, the conclusion avoids clinical claims and treats the system as a research prototype.

5.1 Summary of Contributions

This thesis proposed a therapist-led virtual group therapy prototype for student mental-health support. The system combines DASS-21-based onboarding, route-aware intervention, blackboard coordination, Yalom-inspired peer voices, therapist planning, safety validation, and session memory. Instead of generating a direct chatbot reply, the system first interprets the user’s situation, selects a therapeutic route, controls peer participation, and then produces a final therapist response.

The main contribution is a structured design for controlled group-style support. Nam and Linh are not independent therapists and do not control the session. Their role is limited to short peer contributions that may support universality, catharsis, hope, or interpersonal learning. The therapist core remains responsible for the session flow, safety boundaries, and final response. This makes the system closer to a facilitated virtual group than to a free multi-agent chat.

The second contribution is the VSM evaluation framework. VSM-All provides synthetic multi-turn student-support cases for testing route fidelity, conversation continuity, peer timing, safety boundaries, and overall response quality. The full

benchmark shows that the proposed system performs better than the evaluated baselines, especially in safety, group dynamics, continuity, and structured therapeutic support.

5.2 Limitations

The results should be interpreted carefully. Although the full VSM-All benchmark was conducted, the evidence remains prototype-level because the cases are synthetic and the evaluator is LLM-based. The benchmark is useful for structured comparison and regression testing, but it cannot replace clinician review or real-user evaluation.

The system also has clinical limitations. It has not been tested in a clinical trial, has not been validated by licensed therapists on real student users, and should not be treated as a treatment provider. It may still produce generic, repetitive, poorly timed, or incomplete responses, especially in CBT cases that require careful Socratic questioning and cognitive restructuring.

Finally, the architecture is more complex than a single-prompt chatbot. The blackboard, peer drafts, therapist planner, validators, and memory updater make the system more controllable, but they can also increase latency and token usage. A real deployment would need stronger optimization, monitoring, and human oversight.

5.3 Ethical Boundaries and Future Work

Mental-health support systems require strict boundaries. The prototype should not diagnose users, give medication advice, promise recovery, replace professional care, or encourage emotional dependence on the system. Peer voices also require caution because they can make the interaction feel socially real. For this reason, Nam and Linh must remain limited, transparent, and therapist-governed.

Future work should focus on three directions. First, the benchmark should be validated with human and clinician review so that LLM-judge scores can be calibrated against expert judgment. Second, the CBT route should be strengthened with better stage evidence, better guided-discovery questions, and stricter checks

against premature advice. Third, the system should be optimized for lower latency and safer long-term memory before any real user-facing deployment is considered.

5.4 Conclusion

This thesis does not claim that the system provides clinical treatment. Its contribution is a controllable architecture and evaluation workflow for studying virtual group-style mental-health support. The results suggest that therapist-led coordination, bounded peer participation, and explicit safety checking can make LLM-based support more structured than a generic chatbot response. With clinician audit, real-user studies, and further safety work, the prototype can serve as a useful research platform for student-support dialogue.

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Appendix A

Route Stage Contracts

The system uses stage contracts to keep CBT, MBI, and BA responses aligned with their intended route. These contracts are engineering controls. They are not clinical protocols, but they define what the system should attempt to do at each step.

Table A.0.1: CBT stage contract used by the prototype.

Stage	Therapist goal	Main constraint
Venting	Validate emotion and invite context	Do not challenge thoughts yet.
ABC model	Separate event, automatic thought, and emotion	Do not label distortion too early.
Distortion	Name the cognitive pattern gently	Avoid harsh correction.
Socratic	Ask evidence or double-standard question	Peer agents should usually remain silent.
Reframe/action	Form a balanced thought and small next step	Keep action realistic and specific.

Table A.0.2: MBI stage contract used by the prototype.

Stage	Therapist goal	Main constraint
Grounding	Stabilize attention and body awareness	Do not analyze thoughts.
Decentering	Help the user observe thoughts as mental events	Do not debate whether thoughts are true.
Body scan	Notice physical sensations without judgment	Keep language concrete and slow.
Mindful action	Choose one small present-moment action	Avoid large plans or productivity pressure.

Table A.0.3: BA stage contract used by the prototype.

Stage	Therapist goal	Main constraint
Energy check	Validate depletion and ask for energy level	Do not assign a large task.
Micro-action	Select one tiny action matched to energy	Keep the action small enough to start.
Barrier/schedule	Identify obstacle and set a concrete time or cue	Do not blame the user for avoidance.
Reward/momentum	Reinforce completion and decide rest or next step	Do not escalate too quickly.

The contracts are also used in benchmark design. Each VSM turn contains an expected route, expected stage, required technique hint, and forbidden patterns. The deterministic scorer checks these fields for the proposed system, while external baselines are evaluated from visible text.

Appendix B

Agent Prompting and Safety Boundaries

The prompts are written to keep each agent narrow and auditable. The system does not rely on the prompts alone; validators and guardrails remain responsible for output control. The following appendix summarizes the prompt-level constraints used in the prototype.

B.1 Therapist Orchestrator

The therapist orchestrator is instructed to read the blackboard in a fixed order: risk, route, stage, stage goal, required Yalom factors, peer drafts, and recent history. Before producing the visible response, it forms an internal plan containing the user's immediate need, selected technique, reason for the technique, safety concern, and peer decision.

The therapist must preserve a single responsible clinical voice. If a peer draft matches the current stage and Yalom factor, the therapist may include or paraphrase it. If the draft is repetitive, unsafe, or off-stage, the therapist must discard it. When the required Yalom factor is none, the default decision is therapist-only output.

B.2 Nam: Emotional Mirror

Nam is designed for universality and catharsis. His prompt emphasizes short emotional reflection, shame reduction, and the feeling that the user is not alone. Nam is not allowed to ask CBT technique questions, diagnose, provide advice, or lead the therapeutic flow.

Nam should return no contribution when the current stage requires focused therapist technique. Examples include CBT Socratic questioning, acute MBI grounding, and safety cases. This silence policy is important because group-style support can be harmful if it interrupts stabilization or guided discovery.

B.3 Linh: Hope Witness

Linh is designed for hope and interpersonal learning. Her prompt allows brief lived-experience style statements, but the experience must remain short and realistic. Linh must not promise that the user will recover, tell the user what to do, or turn the conversation toward herself.

Linh is useful when the user expresses hopelessness, asks whether anyone has recovered from a similar situation, or reaches a stage where a small next step can be reinforced. Like Nam, she can return no contribution when peer support is not needed.

B.4 Safety Boundary Statement

The system described in this thesis is a research prototype. It is intended for studying structured AI-supported mental-health dialogue and benchmark design. It is not intended to diagnose, treat, or replace professional care.

When the system detects crisis-like language, it should prioritize immediate safety, encourage contact with trusted people or emergency services, and avoid peer-agent participation. When the user asks for diagnosis, medication dosage, or medical decisions, the system should set a boundary and redirect the user to qualified professionals.

Any real deployment would require additional privacy review, clinical supervision, risk escalation procedures, consent design, data minimization, and institutional accountability.

Appendix C

System and Benchmark Artifacts

This appendix summarizes the implementation and benchmark artifacts used by the thesis. These artifacts support reproducibility of the engineering evaluation, but they do not establish clinical validity.

C.1 System Implementation Notes

The backend exposes chat and streaming routes through the API layer. The dialogue graph is implemented in the AI engine and uses a shared blackboard state. The frontend provides the chat interface, conversation history, model selection, and DASS-21 onboarding form.

The system response format separates internal drafts from visible messages. Peer drafts can be logged for debugging, but only therapist-approved messages are sent to the user. This is important for both safety and benchmark consistency because the conversation history should contain only what the user actually saw.

C.2 Benchmark Artifacts

Each VSM benchmark run writes structured files:

- `per_case_results.jsonl`: raw outputs, metadata, and deterministic scores for each case;
- `failures.jsonl`: hard failures, route mismatches, stage mismatches, peer mismatches, and fallback events;
- `judge_results.jsonl`: judge scores for each turn;
- `score_summary.json`: aggregate benchmark metrics;

- `human_audit_template.csv`: sampled turns for manual review;
- `report.md`: generated benchmark tables and figures.

C.3 Recommended Final Evaluation Procedure

The final benchmark should be run in stages. First, the full Probe split should be run with the proposed system and lightweight baselines. Second, a balanced Session-Core subset should be evaluated with a real LLM judge. Third, 20–30 sampled turns should be manually audited. Only after these steps should the full Session-Core split be used for final reporting.

This staged procedure reduces the risk of spending API budget on a flawed benchmark configuration. It also makes the final report easier to defend because each result can be traced to raw outputs, deterministic checks, judge scores, and manual audit notes.